

# Technology-Mediated Services in Health Care: Evaluating Telehealth Delivery Models for Underserved Populations

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## Abstract

This study examines how telehealth delivery models are associated with access to care and protocol adherence in underserved populations. While prior research reports mixed evidence on telehealth’s effectiveness, we argue that these inconsistencies arise because telehealth is not a single intervention but is implemented through distinct service configurations that address different barriers. Drawing on the theoretical framework of digital economics, we conceptualize telehealth as a technology-mediated service that can alter key economic frictions, including search and verification, transportation, tracking, and setup costs. We focus on two widely adopted models—specialist-facilitated telehealth and remote telehealth—and theorize that each addresses different subsets of these frictions, leading to distinct effects on care access and protocol adherence across settings. Using panel data from U.S. Community Health Centers on mental health services, we document three findings. First, adoption of the specialist-facilitated model is associated with increased access to mental health care, particularly in areas with greater clinical need. Second, remote telehealth is not associated with increased access on average but is associated with higher adherence to standardized depression screening and follow-up protocols. These effects are consistent with improvements in screening and documentation processes rather than increases in utilization. Third, the estimated effects of both models vary across local cost environments. The access effects of specialist-facilitated telehealth are larger in areas where search and verification frictions are greater. In contrast, the effects of remote telehealth are attenuated where patient-side setup costs are high but are larger where transportation barriers and documentation frictions are greater. These findings advance a model-specific perspective on telehealth’s effects, extend the digital economics framework to technology-mediated services, and provide evidence that the estimated effects of digital health interventions vary with how delivery models interact with underlying cost environments.

*Keywords:* Telehealth, Healthcare Access, Community Health Centers, Underserved Population

## 1 Introduction

Digital technologies have dramatically reduced the cost of storage, computation, and data transmission, enabling new forms of technology-mediated service delivery. Across industries, from education and financial services to professional consulting and retail, digital technologies have changed how services

are initiated, coordinated, and delivered. These shifts decouple service provision from physical proximity and enable new forms of interaction between organizations and their users. Healthcare is no exception. Telehealth, the provision of care through digital technologies, is often promoted as a way to expand access to care, particularly for populations facing geographic and logistical barriers. Yet, empirical evidence on the impact of telehealth is mixed. While some studies report improvements in access and outcomes (e.g., Ashwood et al. (2017); Sun et al. (2020)), others find limited or uneven effects (e.g., McCullough et al. (2021); Sun and Wang (2021)). Although this variation is often interpreted as contradictory evidence about telehealth's effectiveness, we argue that it reflects differences in where and how telehealth is implemented. In particular, telehealth is often treated as a single intervention, obscuring meaningful variation in how it is implemented. In practice, it encompasses distinct delivery models that differ in how care is organized, who bears the burden of coordination, and which constraints are relaxed or intensified. As a result, its impact depends on how different delivery models interact with specific organizational and geographic contexts.

We study two widely used telehealth delivery models. In the specialist-facilitated model, patients visit a clinic where providers coordinate virtual consultations with external specialists. In the remote model, patients receive care from non-clinic locations through virtual encounters with providers. We examine how these models affect two core dimensions of healthcare performance: access to care and adherence to clinical protocols. Access captures whether telehealth expands realized utilization among patients facing structural barriers, while protocol adherence reflects the degree to which care follows evidence-based clinical protocols, including screening and follow-up planning, which are important for improving outcomes. To understand the impact of different telehealth delivery models, we extend the economics of digitization framework of Goldfarb and Tucker (2019) and conceptualize telehealth as a technology-mediated service that creates value by altering economic frictions. We focus on four frictions: search and verification costs, transportation costs, tracking costs, and setup costs. We argue that telehealth models are distinct service configurations that influence subsets of these frictions. Outcomes therefore depend on how strongly the frictions alleviated by a given delivery model constrain care delivery in a particular environment.

We study these models in U.S. Community Health Centers (CHCs), which serve low-income and medically underserved populations. Studying telehealth in this setting is particularly important for two reasons. First, CHCs provide a high-friction setting with overlapping barriers to care, making them well suited to test our theoretical framework and to understand how different telehealth models address economic frictions under varying cost environments. Second, understanding whether telehealth expands access for

underserved populations is substantively important, as these populations face the greatest structural barriers to care and thus stand to benefit the most, or lose the most, from digital transformation. We focus on mental health care, where unmet need is high, specialist shortages are persistent, and telehealth adoption has been widespread. More than one in five U.S. adults experiences a mental illness, with higher prevalence in low-income communities (Ridley et al., 2020), yet access remains constrained by provider scarcity (White House Briefing, 2022). This combination of high demand and constrained supply makes CHCs' mental health services a compelling setting for examining how different telehealth models relate to access and protocol adherence under varying constraints.

Using panel data from 2013 to 2019 and a difference-in-differences (DiD) design with staggered adoption, we document three findings. First, adoption of the specialist-facilitated model is associated with higher realized access to mental health care, measured by the number of patients served and total mental health visits. The estimated effects are larger in areas with a higher prevalence of serious mental illness, consistent with the interpretation that the model helps address unmet demand where clinical need is greatest. Second, the remote model is not associated with higher overall mental health utilization on average but is linked to higher adherence to standardized depression screening and follow-up protocols, strengthening a core process foundation for better downstream outcomes. Third, the effects of both models vary systematically with local cost environments. The estimated access effects of specialist-facilitated telehealth are larger in provider-scarce areas (high search and verification costs). For remote telehealth, the estimated access effects are attenuated in areas with low device access (high setup costs) but become larger in areas with low vehicle access (high transportation costs). The estimated effects of the remote model on protocol adherence are larger in areas with greater documentation frictions (high tracking costs). These patterns suggest that the estimated effects of telehealth vary with how delivery models interact with underlying economic frictions.

This study makes three contributions. First, while prior research shows mixed results on telehealth adoption, we shift the focus to delivery models, showing that specialist-facilitated and remote telehealth, as distinct service configurations, produce systematically different outcomes. Second, we extend the digital economics framework to technology-mediated services and provide a theoretical explanation for these varied effects by linking each telehealth model to the specific frictions it may alleviate or intensify. Notably, we identify setup costs, such as access to devices and connectivity, as a key contextual constraint that can attenuate the expected benefits of remote models for underserved populations. Third, unlike prior

studies, which often focus on a single hospital or a single dimension of disadvantage (e.g., rural vs. urban), we focus on underserved populations with multidimensional disadvantages. This distinction is important for understanding whether telehealth can meaningfully expand access in settings where multiple frictions interact. CHCs serve patients facing overlapping financial, insurance, provider, transportation, and digital barriers. This setting allows us to examine how telehealth models perform when multiple frictions are jointly binding, providing a more comprehensive view of telehealth’s role in underserved communities. Our findings can inform health organizations in aligning telehealth strategies with the specific constraints faced by target populations, thereby promoting effective use. More broadly, our findings suggest that the effects of technology-mediated services depend on how specific service configurations interact with the local economic frictions they are designed to alleviate.

## 2 Related Work

Prior research has examined telehealth’s impact on access, utilization, and care delivery, but has produced mixed findings. By reducing travel time and expenses, telehealth can expand access to care (Ashwood et al., 2017; Butzner and Cuffee, 2021) and connect patients to specialists not available locally (Sun and Wang, 2021; Hwang et al., 2022). However, disparities in digital literacy and access to technology can limit participation in virtual care (McCullough et al., 2021; Zhou et al., 2024). Empirical evidence reflects this tension.

[Insert Table 1]

As shown in Table 1, prior research has examined outcomes including utilization, spending, operational efficiency, and access disparities. Studies find that telehealth can increase utilization rather than substituting for in-person care (Ashwood et al., 2017), increase visit incompleteness due to patient abandonment under scheduling delays (Qin et al., 2025), and change channel usage and follow-up patterns (Lekwijit et al., 2026; Ayabakan et al., 2024). Evidence on access equity is also mixed. Telehealth use can be concentrated in affluent areas with strong broadband infrastructure (McCullough et al., 2021) and may not reduce rural-urban disparities (Sun and Wang, 2021), while other studies find increased visit rates in resource-constrained settings (Delana et al., 2023) or increased connections between providers and underserved patients (Hwang et al., 2022). Collectively, these findings suggest that telehealth’s effects are heterogeneous, potentially reflecting differences in implementation and context.

Much of the existing literature treats telehealth as a single intervention. In practice, however, telehealth

is implemented through different delivery models, most commonly specialist-facilitated and remote models. Figure 1 illustrates these two models. The specialist-facilitated model links two sites—the client site, where a provider is physically present with the patient, and the specialist site, where a consulting specialist is located (Elrod and Fortenberry Jr, 2017). This model is typically used after initial screening and diagnosis, when patients require specialty care unavailable within their usual health organizations. In contrast, the remote model enables patients to engage with providers through virtual encounters from home or other non-clinic locations. By allowing clinical encounters to occur remotely, this model offers greater flexibility but places more responsibility on patients to initiate and maintain digital interactions. These differences suggest that telehealth’s effects may vary across delivery models and the local conditions under which each model is implemented. Yet existing research provides limited theoretical guidance on how and why these effects vary across settings.

[Insert Figure 1]

Additionally, prior research on access equity often defines underserved populations along a single dimension, such as rurality, income, or broadband access. This approach overlooks the multiple, overlapping constraints underserved populations frequently face. Reliance on commercial insurance or hospital data may also underrepresent structurally constrained patients and limit our understanding of telehealth in settings where multiple barriers are simultaneously binding.

Our work addresses these gaps in three ways. First, we distinguish between telehealth delivery models, helping to explain mixed findings in prior studies. Second, we examine two performance dimensions, access and protocol adherence. Access captures whether telehealth expands reach to patients facing structural barriers, while protocol adherence reflects whether care follows clinical guidelines and workflow standards. Third, we study CHCs, which provide care to underserved populations facing overlapping financial, provider, transportation, and digital constraints. This high-need, resource-scarce setting allows us to examine how the effects of different telehealth models vary across levels of structural barriers.

### **3 Theory Framework and Hypothesis Development**

Digital economics theory examines how digitization creates value by reducing economic frictions. Goldfarb and Tucker (2019) identify five core frictions: search costs (looking for information), verification costs (confirming identity and trust), transportation costs (delivering services across distance), tracking costs (connecting people or goods with relevant data), and replication costs (reproduction). This

framework is particularly relevant to fully digital goods and services—such as media, software, and online platforms—where transactions are non-rivalrous, and marginal costs approach zero.

However, many services that adopt digital technologies are not fully digital but technology-mediated (Froehle and Roth, 2004). In these settings, interactions occur through digital interfaces, but service delivery continues to rely on human providers whose activities must be coordinated (Barrett et al., 2012). Related Information Systems research on process virtualization similarly recognizes that shifting physical processes to virtual environments depends on the characteristics and requirements of the underlying process (Overby, 2008). Telehealth exemplifies this class of services. Although encounters are virtual, care delivery still depends on licensed professionals, coordinated workflows, and patient engagement. We extend the digital economics framework to technology-mediated services by conceptualizing telehealth models as distinct service configurations that differentially alter economic frictions. We focus on four frictions central to healthcare delivery. *Search and verification costs* capture the effort required to identify appropriate providers and confirm their availability and suitability. We combine search and verification costs because identifying appropriate healthcare providers and confirming their qualifications are closely intertwined and typically occur through the same processes (e.g., referrals, credentialing, insurance), making them difficult to separate. *Transportation costs* capture the burden of physical travel, while *tracking costs* capture frictions in recording, storing, and linking information across visits. We exclude replication costs because technology-mediated services remain rivalrous and instead introduce *setup costs*, which capture the technological and cognitive requirements of initiating digital interactions, including device access, connectivity, and digital literacy. The performance implications of telehealth therefore depend on how each delivery model alters these frictions and where they are more or less binding.

Table 2 summarizes the expected change in each friction relative to conventional in-person care. In our setting, search and verification, transportation, and setup costs are primarily patient-side frictions affecting access and care-seeking, whereas tracking costs arise from provider-side documentation and information processes, with indirect consequences for patients.

[Insert Table 2]

**Specialist-Facilitated Telehealth.** In the specialist-facilitated model, patients visit the CHC, where providers coordinate virtual consultations with external specialists.<sup>1</sup> This configuration shifts the burden

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<sup>1</sup>Mental health specialists refer to licensed behavioral health providers, including psychiatrists, clinical psychologists, and other licensed behavioral health providers (e.g., clinical social workers, counselors, and marriage and family therapists)—consistent with the classification used by the Health Resources and Services Administration (HRSA).

of identifying, verifying, and coordinating specialist care from patients to the CHC, reducing *search and verification costs*. It also reduces *transportation costs* by allowing patients to receive specialist care at their local CHC rather than traveling to specialty facilities. By hosting the visit, the CHC shields patients from additional *setup costs* associated with personal device access, connectivity, and digital literacy. Because this model is typically used after initial screening and diagnosis within the CHC, the intake and documentation processes that give rise to tracking costs remain largely unchanged relative to in-person care. We therefore do not expect it to meaningfully affect tracking costs.

In safety-net settings characterized by high unmet need and provider shortages, reducing these patient-side frictions can increase the likelihood that unmet mental health needs translate into completed visits. Accordingly, we hypothesize:

**Hypothesis 1.** CHC adoption of the specialist-facilitated telehealth model is associated with greater realized access to mental health services.

**Remote Telehealth.** The remote model enables patients to connect with providers from their own locations. This configuration eliminates *transportation costs* associated with clinic visits but introduces patient-side *setup costs* related to device access, connectivity, and digital literacy. It leaves search and verification burdens largely unchanged relative to in-person CHC visits because remote delivery does not necessarily alter how patients identify and access providers. Because setup barriers may offset transportation savings, the net effect of the remote model on access is theoretically ambiguous and remains an empirical question. We therefore do not advance a directional hypothesis regarding access.

The remote model may, however, influence documentation, monitoring, and follow-up by shifting a greater share of these activities into digitally supported workflows. In traditional in-person mental health visits, paper-based screening followed by manual scoring and EHR entry creates a physical-to-digital handoff that can contribute to incomplete records and omitted steps, elevating *tracking costs* (Sattler et al., 2024; Kawado et al., 2003; Hwang et al., 2022). In these cases, patients bear the downstream consequences, such as missed screenings or delayed follow-up care (Callen et al., 2012).

Remote telehealth can reduce these tracking costs by supporting more structured documentation and follow-up workflows (Haleem et al., 2021; Khairat et al., 2025). Screening and follow-up may be incorporated through electronic tools or provider-supported documentation (O’Callaghan et al., 2022). For example, patients may complete screening forms electronically before an encounter, facilitating subsequent documentation and follow-up. To the extent that remote delivery supports these processes, it can reduce

tracking costs and the risk of missed protocol steps, although implementation may vary across CHCs.

Because adherence to screening and follow-up protocols is measured at an early stage of mental healthcare, it is less relevant to the specialist-facilitated model, which is typically deployed after initial screening and diagnosis. Accordingly:

**Hypothesis 2.** CHC adoption of the remote telehealth model is associated with higher adherence to clinical protocols.

**Heterogeneity Across Cost Environments.** Because telehealth models alter different economic frictions, their effects should vary with local cost environments. We expect larger effects where the frictions a model alleviates are more severe and smaller effects where other constraints remain binding. Specialist-facilitated telehealth should yield larger access gains where provider scarcity and search and verification costs are high. Remote telehealth access effects should be attenuated where device access is limited but larger where transportation barriers are greater. Its protocol-adherence effects should likewise be larger where documentation and tracking frictions are greater. Section 5.3 tests these predictions using the empirical proxies summarized in Table 2 to characterize local *cost environments*.

## 4 Setting and Empirical Approach

### 4.1 Telehealth Implementation in Health Centers

CHCs are nonprofit, community-based organizations established under the Public Health Service Act to deliver comprehensive primary care in medically underserved areas regardless of patients' ability to pay (NACHC, 2024). In 2022, they served 30.2 million patients in the U.S.; 50% were covered by Medicaid, 19% were uninsured, and 45% were living at or below the Federal Poverty Level (NACHC, 2024). These characteristics distinguish CHCs from hospital-based and privately insured settings, making them a resource-constrained context for examining care delivery interventions. By 2019, 42.7% of CHCs reported using telehealth, with mental health as the most common application (75.3% among adopters).

Mental health services in CHCs are delivered through a combination of on-site behavioral health clinicians and external specialty partnerships (Shin et al., 2013), with limited specialist availability in many service areas. As in much of the U.S., mental health care is often delivered in primary care settings (Pettersson et al., 2014), a pattern especially pronounced in CHCs due to specialist shortages: approximately 20% of CHCs employ no mental health specialists, and nearly 90% have fewer than 0.5 specialists per 1,000 patients. These shortages limit access to specialty care and continuity of follow-up, which is critical for

timely intervention and patient health (Silverman, 2024).

Given patients' transportation barriers, insurance instability (affecting search and verification costs), and uneven broadband access, alongside CHCs' staffing and financial constraints (Cole et al., 2018; Swenson and Ghertner, 2020), this setting features substantial frictions. Mental health care in CHCs is therefore particularly sensitive to changes in care delivery processes, allowing us to test our theoretical framework and understand the heterogeneous effects of different telehealth models on access and protocol adherence within a multidimensional cost environment. This setting also allows us to evaluate whether telehealth expands access for underserved populations.

## 4.2 Data Sources and Sample Selection

We use Uniform Data System (UDS) data from annual reports submitted to the Health Resources and Services Administration (HRSA), which administers the Health Center Program. The data provide detailed information on CHC locations, funding, patient and visit volumes, and telehealth practices. These reports are mandatory and undergo structured review, and CHCs' reliance on HRSA funding creates strong incentives for accurate reporting (Maccini and Davis, 2021). We construct a panel dataset covering 2013 to 2019, excluding data from 2020 onward to avoid confounding from the COVID-19 pandemic. We supplement these data with county-level measures of device and vehicle access from the American Community Survey and measures of mental health provider availability from the National Provider Identifier Registry.

To identify telehealth adoption, we use telehealth implementation information from the UDS, which was first collected in 2016. We define the treatment group as CHCs that adopted mental health telehealth between 2017 and 2019 and continued its use throughout the sample period, and the control group as CHCs that did not adopt mental health telehealth during the study period. Synchronous, audio-visual, direct-to-patient visits were the predominant modality during this time. We exclude early adopters in 2016 due to unobserved adoption timing and apply additional sample restrictions described in Table 3, resulting in a final sample of 964 CHCs.

[Insert Table 3]

## 4.3 Measures

We examine two dimensions of performance: access to care and protocol adherence. Access is measured using the total number of unique mental health patients ( $MentalPatients_{i,t}$ )<sup>2</sup> and the total number of mental

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<sup>2</sup>This captures the number of unique patients receiving care for mental health conditions, including depression, mood disorders, anxiety disorders, attention deficit disorders, substance abuse disorders, and other mental disorders.

health visits ( $MentalVisits_{i,t}$ ), measured at each CHC  $i$  in year  $t$ . We report both unique patient counts and total visit volumes because they capture complementary aspects of care utilization. Unique patient counts reflect expansion in realized access, whereas total visits capture subsequent utilization among patients who have entered the system. Together, these outcomes provide a more complete picture of the system-level consequences of telehealth adoption and distinguish expansion in patients served from changes in care intensity among existing patients. An increase in visit frequency through improved follow-up or convenience does not necessarily indicate improved access at the population level, whereas an increase in patients served indicates that the system is reaching additional individuals.

We use total patient counts and visit volumes across both modalities to measure access for two reasons. First, telehealth is integrated into the overall care delivery system rather than operating as a standalone channel. In practice, telehealth adoption influences both virtual and in-person visits, with care delivered through a combination of modalities (e.g., initial screening may occur in person, followed by virtual follow-up visits). As a result, telehealth can both substitute for and complement in-person care. Measuring utilization by modality alone cannot distinguish between these effects. Total utilization therefore captures the net system-level change in care use following telehealth adoption, complementing the unique-patient measure of realized access. Second, consistent with Andersen’s concept of realized access (Andersen, 1995), increases in total utilization in underserved settings are more likely to reflect the release of latent demand rather than supply-induced demand (i.e., over-treatment). In Section 5.1, we provide supporting evidence.

Protocol adherence ( $ProtocolAdherence_{i,t}$ ) is measured following HRSA guidelines as the percentage of patients aged 12 years and older who are screened for depression and, if screening positive, have a follow-up plan documented on the same day. We examine protocol adherence because it reflects the reliability with which evidence-based clinical processes are executed. Adherence to standardized processes is essential for translating patient encounters into appropriate care, as failures at these stages can lead to missed diagnoses or delayed treatment, negatively affecting health outcomes. Compared to clinical outcomes, protocol adherence more directly captures process execution and is well-suited for our setting, where telehealth—especially the remote model—may reshape documentation, monitoring, and follow-up through more digitally supported workflows.

We distinguish between specialist-facilitated and remote models for mental health services. CHCs may also choose not to adopt telehealth, in which case services are delivered exclusively in person. We code

CHCs' telehealth models using Health Center Data on reported mental health telehealth practices. Detailed coding rules and survey items are provided in Online Appendix A.

We exclude the very small subset of observations that report adopting both models (2.9% of CHC-year observations). Because our framework treats delivery models as distinct configurations that alter different frictions, restricting the sample to single-model adopters yields cleaner identification and a more direct test of our theoretical predictions. Given the structure of UDS reports, when a CHC reports adopting both models and multiple practice areas, it is not possible to determine whether both are used for mental health services, introducing ambiguity in treatment assignment. Including such observations would introduce measurement error and confound attribution of effects to a particular configuration. We therefore focus on centers adopting a single model.

We construct binary indicators for each telehealth delivery model. Within the corresponding estimation sample, *SpecialistFacilitatedModel*<sub>*i,t*</sub> equals 1 in all years after adoption of the specialist-facilitated model for mental health services and 0 in all years prior to adoption. *RemoteModel*<sub>*i,t*</sub> is defined analogously within its corresponding estimation sample. These variables capture organization-level adoption rather than visit-level use. Estimates should therefore be interpreted as organizational-level adoption effects rather than visit-level treatment-on-the-treated effects (Angrist et al., 1996; Imbens and Rubin, 2015).

We include time-varying control variables capturing organizational capacity, behavioral health emphasis, patient composition, and local conditions. These include EHR adoption, prior clinical quality,<sup>3</sup> service scope, number of service sites, behavioral health funding, and telehealth use for non-mental health services; patient composition (shares of homeless and Medicaid patients) and local digital device access; and service-area characteristics, such as the share of CHCs' service delivery sites located in rural areas and local mental health specialist supply. Table A1 provides detailed variable definitions.

## 4.4 Empirical Strategy

### 4.4.1 Estimating the Telehealth Implementation Effects

To examine changes in outcomes associated with telehealth adoption, we employ a DiD approach with staggered treatment timing. Our identification strategy exploits variation in the timing of telehealth adoption across CHCs, comparing changes in outcomes before and after adoption for treated centers relative to those

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<sup>3</sup>We measure quality using Community Health Quality Recognition (CHQR) indicators for childhood immunization, cervical cancer screening, pediatric weight assessment/counseling, and BMI screening with follow-up. CHCs ranking in the top quartile across all four measures in the previous year are classified as top-quality performers.

that do not adopt during the study period. To improve comparability between adopters and non-adopters, we use a propensity score-matched sample constructed using pre-treatment covariates for all DiD estimations. Detailed matching procedures are provided in Online Appendix C. We estimate the following model separately for specialist-facilitated and remote telehealth:

$$Y_{i,t} = \beta \cdot \text{SpecialistFacilitatedModel}_{i,t} + \mathbf{X}_{i,t}\gamma + \alpha_i + \delta_t + \varepsilon_{i,t}, \quad (1)$$

$$Y_{i,t} = \beta \cdot \text{RemoteModel}_{i,t} + \mathbf{X}_{i,t}\gamma + \alpha_i + \delta_t + \varepsilon_{i,t}. \quad (2)$$

Each specification compares CHCs adopting one telehealth delivery model with CHCs that never adopt mental health telehealth during the study period. CHCs adopting the other delivery model, or both models, are excluded from that specification’s estimation sample. The dependent variable,  $Y_{i,t}$ , represents the outcome of interest:  $\text{MentalPatients}_{i,t}$ ,  $\text{MentalVisits}_{i,t}$ , or  $\text{ProtocolAdherence}_{i,t}$ . The first two dependent variables are log-transformed due to their skewed distributions.  $\text{SpecialistFacilitatedModel}_{i,t}$  and  $\text{RemoteModel}_{i,t}$  are our main independent variables as defined in Section 4.3. The parameter of interest,  $\beta$ , captures the estimated change in the outcome associated with telehealth adoption under each delivery configuration.  $\mathbf{X}_{i,t}$  represents a vector of control variables. CHC fixed effects ( $\alpha_i$ ) account for all time-invariant differences across health centers, and year fixed effects ( $\delta_t$ ) capture common time trends affecting all CHCs.  $\varepsilon_{i,t}$  is the error term with standard errors clustered at the CHC level.

**Addressing staggered adoption and endogeneity concerns.** We conduct robustness checks addressing staggered adoption and endogeneity. We estimate event study models and aggregate them using a stacked design that compares each treated cohort (e.g., CHCs that adopt telehealth in 2018) with never-treated controls. Identification is further supported by evidence of parallel pre-trends, placebo timing permutations, and an instrumental variable approach with consistent results. Please refer to Section 5.4 and Online Appendix E for these robustness tests.

#### 4.4.2 Heterogeneity Analysis by Cost Environment

We next examine whether the estimated effects of telehealth models vary across cost environments that reflect differences in economic frictions and influence care access and delivery. Our framework predicts larger estimated effects in environments where the frictions a model alleviates are more severe, and smaller estimated effects where other constraints remain binding. Guided by our framework, we focus on four cost dimensions: search and verification costs, setup costs, transportation costs, and tracking costs. For each

dimension, we interact telehealth adoption with a corresponding indicator for high-cost environments for CHC  $i$ , calculated using pre-adoption measures of local conditions, and equal to one if the corresponding cost proxy is higher than the sample median. We operationalize these cost proxies as binary indicators because our theoretical framework predicts differences in the estimated effects of telehealth between high- and low-friction environments rather than a linear pattern of effect heterogeneity across the full range of friction intensity. The empirical specification extends Equations (1) and (2) by including interaction terms:

$$Y_{i,t} = \beta_1 \cdot \text{TelehealthModel}_{i,t} + \beta_2 (\text{TelehealthModel}_{i,t} \times \text{HighCost}_i) + \mathbf{X}_{i,t}\gamma + \alpha_i + \delta_t + \varepsilon_{i,t}. \quad (3)$$

Here,  $\beta_1$  captures the estimated change associated with telehealth adoption in low-cost environments, while  $\beta_2$  captures the differential change in high-cost settings. All models include CHC and year fixed effects, with standard errors clustered at the CHC level.

We proxy search and verification costs using local specialist scarcity, measured by the psychiatrist-to-population ratio, as psychiatrists are among the scarcest and most specialized mental health providers. Psychiatrist scarcity is not itself a search and verification cost, but it reflects local environments in which patients face greater difficulty identifying and obtaining specialty mental health care. This measure characterizes the conventional local specialty-care environment rather than the geographic location of specialists providing telehealth consultations, who may be located outside the CHC service area. Using this proxy, we examine whether the specialist-facilitated model has larger access effects in provider-scarce areas, as shifting coordination from patients to CHCs can alleviate these constraints. Setup costs are proxied by lack of local device ownership, measured as the share of households without a desktop, laptop, smartphone, or tablet, capturing patient-side digital capacity relevant to the remote model. We examine whether its effects attenuate in areas with low device ownership, where digital access constraints are more binding. Transportation costs are measured using the share of households without a vehicle. To ensure this proxy reflects travel burdens rather than public transit infrastructure, we exclude CHCs located in the top 5% of areas by residential density. We examine whether the remote model has larger access effects in areas with low vehicle ownership, where transportation costs are greater, as the remote model can eliminate travel. Finally, we proxy tracking costs using the pre-adoption share of patients with missing income information in UDS reports. Income information is routinely collected by CHCs to determine patient eligibility for the HRSA-required Sliding Fee Discount Program and is therefore part of standard intake and documentation

procedures.

Because income information is routinely collected as part of eligibility determination and reporting, persistent differences in missing income rates across CHCs are informative about differences in documentation processes and information management capacity, making this measure a useful proxy for provider-side tracking and recording frictions. We validate this interpretation by confirming that high- and low-tracking=cost CHCs do not differ in baseline protocol adherence or exhibit differential pre-treatment trends (see Online Appendix Tables A23–A24). We test whether remote telehealth’s effect on protocol adherence is larger in these settings, where there is greater scope for improvement in documentation and monitoring processes. Table 4 summarizes the theoretical rationale and empirical implementation for each cost dimension.

[Insert Table 4]

## 5 Results

### 5.1 Telehealth Implementation and Access to Mental Health Care

Table 5 reports the estimated effects of telehealth adoption on realized access and utilization, measured using the number of mental health patients and total mental health visits, respectively. Full regression results are reported in Online Appendix Tables A5 and A6.

The coefficients of 0.116 and 0.182 correspond to approximately 12% and 20% higher numbers of mental health patients and visits, respectively, following adoption of the specialist-facilitated model. These results are consistent with greater realized access and utilization following adoption. As shown in Column 3, specialist-facilitated telehealth is not associated with a significant change in protocol adherence ( $\beta = 0.002$ ,  $p > 0.10$ ), consistent with our theoretical focus on access for this delivery model.

[Insert Table 5]

These increases are consistent with expanded access rather than overutilization. The effects are larger in areas with a higher prevalence of serious mental illness and major depressive episodes (Online Appendix Table A7), suggesting the activation of latent demand where clinical need is greatest. We also find higher penetration within existing service areas following adoption, but no evidence of expansion into new areas. Finally, decomposition analysis shows that the increase in visits is driven primarily by expansion in the number of patients served, with a smaller contribution from higher visit intensity (Online Appendix D). These findings are consistent with our theoretical framework. By reducing search and verification costs

through centralized coordination and lowering transportation barriers while shielding patients from setup constraints, the specialist-facilitated model enables patients facing multiple barriers to translate unmet need into realized visits.

In contrast, we find no evidence that the remote model increases access to mental health care on average. The estimated effects on both patients and visits are small and not statistically significant. This differs from evidence in general or privately insured populations, suggesting that access gains observed in those settings may not generalize to safety-net contexts. In communities with limited digital access, requirements related to devices, connectivity, and private space may constrain uptake, offsetting transportation savings (Chunara et al., 2021).

## **5.2 Telehealth Implementation and Protocol Adherence in Mental Health Care**

As Column 3 in Table 6 shows, adoption of the remote model is associated with a statistically significant increase in protocol adherence, supporting Hypothesis 2. This finding suggests that greater digital mediation may facilitate more structured screening and follow-up documentation. To clarify the mechanism, we decompose protocol adherence into its numerator and denominator in Online Appendix Table A11. Remote telehealth adoption does not change the number of patients eligible for screening but is associated with an increase in patients who receive screening with a follow-up plan documented when screening is positive. These findings are consistent with our argument that shifting a greater share of care activities into digitally mediated workflows may reduce tracking frictions associated with manual handoffs and fragmented documentation. Section 5.3 examines whether these gains are larger where tracking and documentation constraints are greater.

The specialist-facilitated model is not associated with a significant change in protocol adherence, consistent with early-stage screening and documentation occurring within the CHC prior to specialty referral.

[Insert Table 6]

## **5.3 Heterogeneous Effects Across Cost Environments**

We next examine whether telehealth effects vary across local cost environments as predicted by our framework. Table 7 shows that the specialist-facilitated model has no detectable effect on utilization in provider-abundant environments. In areas with limited specialist availability, however, the interaction term is positive and statistically significant, indicating that access gains are concentrated in provider-scarce areas,

consistent with larger estimated effects where search and verification costs are high.

[Insert Table 7]

For the remote model, effects vary with setup costs (Table 8) and transportation costs (Table 9). In areas with relatively high device access, remote telehealth modestly increases utilization, but these gains attenuate sharply in low-device environments (i.e., high setup costs), where the interaction term is negative and statistically significant. This is consistent with digital barriers constraining remote care. By contrast, estimated visit gains are larger in areas with limited vehicle ownership (i.e., high transportation costs). Thus, the estimated effects of the remote model are larger in high transportation-friction environments, but transportation benefits may be insufficient when digital access constraints remain binding.

[Insert Tables 8, 9]

Finally, Table 10 shows that higher protocol adherence associated with remote telehealth is concentrated in centers with lower documentation and tracking capacity (i.e., high tracking costs). Where such capacity is stronger, the effect is negligible. This pattern is consistent with greater scope for improvement in documentation and monitoring processes in high tracking-friction environments.

[Insert Table 10]

Event-study specifications support the parallel trends assumption across the heterogeneity analyses. Overall, each delivery configuration exhibits larger estimated gains in environments where the friction it targets is more severe.

## **5.4 Alternative Model Specifications and Robustness Tests**

We evaluate the sensitivity of our findings to alternative specifications, measurement choices, sample restrictions, and potential unobservable confounding. Results are summarized in Table 11, with details in Online Appendix E.

[Insert Table 11]

## **6 Discussion and Conclusion**

This study shows that telehealth's effects depend on delivery architecture rather than adoption alone. Distinguishing between specialist-facilitated and remote telehealth, we find systematically different outcomes. Specialist-facilitated telehealth is associated with higher numbers of mental health patients served and total visits, particularly in areas with greater unmet demand. In contrast, remote telehealth is not associated with higher utilization on average but is associated with higher adherence to early-stage screening

and follow-up protocols. These findings help reconcile mixed evidence in prior literature, which often treats telehealth as a homogeneous intervention. More broadly, specialist-facilitated telehealth is linked primarily to higher realized access, whereas remote telehealth is linked primarily to improved protocol adherence.

These effects vary systematically across local conditions. The estimated access effects of specialist-facilitated telehealth are larger in areas with limited specialist supply, consistent with the interpretation that the model alleviates search and verification constraints. For remote telehealth, protocol adherence effects are concentrated in areas with greater tracking frictions, whereas access effects are attenuated where patient-side setup costs are high but are larger where transportation frictions are greater. These patterns also have implications for access equity. Specialist-facilitated telehealth may help narrow disparities arising from unequal local specialist availability. In contrast, remote telehealth may reinforce disparities when patients facing high setup costs are less able to benefit from the access it provides, even as the model reduces transportation barriers. Thus, the equity consequences of telehealth depend not only on which existing barriers a delivery model reduces, but also on whether it introduces new barriers to accessing care.

Our study extends the economics of digitization framework beyond fully digital goods to technology-mediated services, where service delivery remains rivalrous and capacity-constrained. We treat economic frictions not as directly observed mechanisms, but as underlying constraints that affect the performance of different delivery models across environments. This perspective shifts the focus from asking whether digital technologies improve performance on average to understanding which service configurations are effective under which local constraints.

Our theoretical framework and empirical findings have important scope conditions. The framework is applicable to settings in which one or more of the economic frictions we theorize are present; applying it requires considering which frictions are salient and how the technology-mediated service alters them. The generalizability of our empirical findings depends on the combination of frictions present in the setting. Our findings are most likely to generalize to healthcare settings resembling U.S. Community Health Centers, where multiple economic frictions—including provider scarcity, transportation barriers, digital access constraints, and documentation challenges—coexist and technology-mediated services can affect several of these frictions simultaneously.

Within these settings, our findings carry implications for policy and organizational strategy. Telehealth adoption should not be treated as a uniform objective, because different delivery models address different

constraints and generate distinct outcomes. Policies that promote telehealth without regard for configuration or context are therefore likely to yield heterogeneous results. In settings characterized by multiple overlapping frictions, aligning technology-mediated service configurations with locally salient constraints may improve the benefits of adoption.

This study has several limitations. First, we do not examine cases where CHCs implement both specialist-facilitated and remote telehealth. Few CHCs adopt both models, and when they do, the data do not allow us to identify which practice areas each model serves. Future research using more granular service-line data could examine how hybrid strategies shape access and cost dynamics. Second, our data do not distinguish between virtual and in-person encounters at the visit level. Although our analysis estimates system-level changes associated with telehealth adoption, we cannot observe visit-level mechanisms or examine substitution and complementarity between virtual and in-person care. Third, although our empirical strategy combines propensity score matching, fixed effects, time-varying controls, and multiple robustness analyses to mitigate selection concerns, residual confounding from unobserved time-varying organizational changes cannot be completely ruled out. Future research using richer longitudinal or experimental data could further strengthen causal identification. More broadly, future work could extend our framework to other technology-mediated services to better understand how digital configurations interact with local constraints in different domains.

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## 7 Tables and Figures

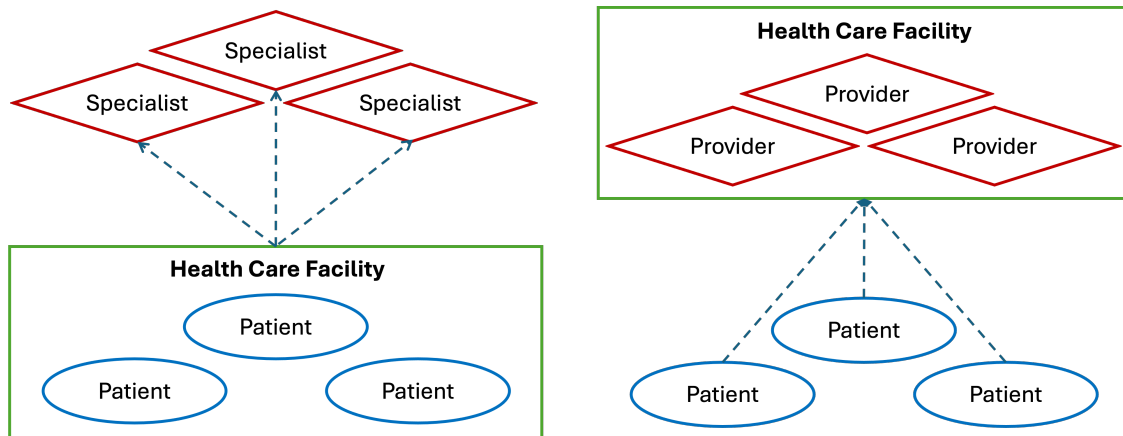


Figure 1: Specialist-facilitated (Left) and Remote Telehealth (Right) Models

Table 1: Empirical Studies on Telehealth

| Study                    | Empirical Context                               | Telehealth model | Primary outcomes                  | Main Findings                                                                                                                                                                                                                                                                        |
|--------------------------|-------------------------------------------------|------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ashwood et al. (2017)    | Commercial insurer claims data                  | R model          | Utilization, spending             | Most telehealth appointments represent new utilization. Spending per enrollee was higher among telehealth users than non-users.                                                                                                                                                      |
| Sun et al. (2020)        | All-payer ED visits                             | S model          | ER efficiency (LOS, waiting time) | Adoption of ER telehealth reduces average patient LOS and wait times.                                                                                                                                                                                                                |
| McCullough et al. (2021) | Commercial insurer claims data                  | Not specified    | Access disparities                | Telehealth use and in-person care were more common among patients in affluent areas with higher income and broadband access.                                                                                                                                                         |
| Sun and Wang (2021)      | Hospital claims, urban vs. rural patients       | Not specified    | Rural–urban access disparities    | Telehealth expansion did not reduce rural–urban disparities in total visits.                                                                                                                                                                                                         |
| Hwang et al. (2022)      | Teleconsultation platform data                  | R model          | Geographic healthcare disparity   | Telehealth connects physicians in resourceful regions with underserved patients, but social, informational, and geographic frictions limit virtual flows.                                                                                                                            |
| Delana et al. (2023)     | Telehealth centers in Southern India            | S model          | Access and utilization            | Network visit rates increased following the introduction of telemedicine centers.                                                                                                                                                                                                    |
| Lekwijit et al. (2026)   | Academic healthcare system                      | R model          | Video and in-person channel usage | Telehealth introduction changed demand patterns across channels.                                                                                                                                                                                                                     |
| Qin et al. (2025)        | Outpatient clinic appointments                  | R model          | Visit incompleteness              | Telehealth visits were more likely to be missed due to delays, primarily because of patient abandonment.                                                                                                                                                                             |
| Ayabakan et al. (2024)   | Insured patient claims data                     | Not specified    | Utilization                       | Telehealth visits reduced follow-up outpatient visits and associated costs.                                                                                                                                                                                                          |
| Zhou et al. (2024)       | National provider data; urban and rural markets | Not specified    | Geographic competition            | Telehealth expansion shifted patient spending from rural to urban providers.                                                                                                                                                                                                         |
| This study               | Underserved population in safety-net CHCs       | S model, R model | Access and protocol adherence     | Specialist-facilitated telehealth is associated with higher access, especially where provider scarcity and unmet need are high. Remote telehealth is associated with higher protocol adherence but not utilization on average. Estimated effects vary with local economic frictions. |

*Notes:* ED = Emergency Department; ER = Emergency Room; LOS = Length of Stay; S model = Specialist-facilitated telehealth model; R model = Remote telehealth model.

Table 2: The Influence of Telehealth Models on Economic Frictions

|                 | <b>Search and Verification</b> | <b>Transportation</b> | <b>Tracking</b>            | <b>Setup</b>  |
|-----------------|--------------------------------|-----------------------|----------------------------|---------------|
| S model         | ↓                              | ↓                     | —                          | —             |
| R model         | —                              | ↓↓                    | ↓                          | ↑             |
| Empirical proxy | Specialist scarcity            | Vehicle ownership     | Missing income information | Device access |

*Note.* S model = Specialist-facilitated telehealth model; R model = Remote telehealth model. Arrows indicate the expected change in each economic friction relative to conventional in-person care; a dash (—) indicates no expected change.

Table 3: Sample Selection Criteria

| <b>Sample selection procedure</b>                                                                                                        | <b>Remaining Number of CHCs</b> |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| All Community Health Centers in HRSA.                                                                                                    | 1,339                           |
| Step 1: Exclude CHCs located outside the 50 U.S. states (i.e., located in U.S. territories)                                              | 1,306                           |
| Step 2: Exclude CHCs with an unknown year of adoption (i.e., those reporting telehealth adoption in 2016, the first year of the survey). | 1,060                           |
| Step 3: Exclude CHCs that marked telehealth adoption but changed later to not adopted.                                                   | 964                             |

Table 4: Cost Dimensions and Empirical Tests for Heterogeneity Analysis

|                       | Search/Verification costs                                                                                               | Setup costs                                                                                                                                  | Transportation costs                                                                                             | Tracking costs                                                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theory</b>         | S model primarily reduces search costs by shifting specialist identification and coordination from patients to the CHC. | R model introduces patient-side setup costs (devices, broadband, digital literacy), which can become binding when digital access is limited. | R model substantially reduces transportation costs by allowing patients to receive care from their own location. | R model reduces tracking costs by shifting a greater share of documentation and follow-up activities into digitally supported workflows, reducing reliance on paper-to-digital handoffs. |
| <b>Rationale</b>      | Search costs are higher in areas where mental health specialists are scarce.                                            | Setup costs are higher in areas where device access and digital readiness are low.                                                           | Transportation costs are higher in areas where vehicle ownership is limited (We exclude metropolitan areas).     | Tracking costs are higher in CHCs where documentation and administrative workflows are fragmented.                                                                                       |
| <b>Empirical test</b> | Interact S model indicator with baseline provider scarcity.<br>Outcome: access                                          | Interact R model indicator with baseline device access.<br>Outcome: access                                                                   | Interact R model indicator with baseline vehicle access.<br>Outcome: access                                      | Interact R model indicator with baseline tracking friction.<br>Outcome: protocol adherence                                                                                               |

Notes: S model = Specialist-facilitated telehealth model; R model = Remote telehealth model.

Table 5: Effects of Specialist-Facilitated Telehealth on Mental Health Access and Protocol Adherence

|                                    | (1)<br><i>Ln(MentalPatients)</i> | (2)<br><i>Ln(MentalVisits)</i> | (3)<br><i>ProtocolAdherence</i> |
|------------------------------------|----------------------------------|--------------------------------|---------------------------------|
| <i>SpecialistFacilitated Model</i> | 0.116**<br>(0.051)               | 0.182***<br>(0.066)            | 0.002<br>(0.026)                |
| Controls                           | Yes                              | Yes                            | Yes                             |
| CHC F.E.s                          | Yes                              | Yes                            | Yes                             |
| Year F.E.s                         | Yes                              | Yes                            | Yes                             |
| Observations                       | 1189                             | 1189                           | 1038                            |
| Adjusted R <sup>2</sup>            | 0.880                            | 0.855                          | 0.600                           |

Notes: Time-varying controls include the local mental health specialist-to-population ratio, a prior-year top-quality indicator, the percentage of Medicaid patients, the percentage of homeless patients, indicators for provision of pediatric and dental services, electronic health record adoption, telehealth use in other service lines, rural location, number of service sites, receipt of behavioral health grants, and the local percentage of households with computing device access. Standard errors are clustered at the CHC level. F.E.s = fixed effects.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 6: Effects of Remote Telehealth on Mental Health Access and Protocol Adherence

|                         | (1)<br><i>Ln(MentalPatients)</i> | (2)<br><i>Ln(MentalVisits)</i> | (3)<br><i>ProtocolAdherence</i> |
|-------------------------|----------------------------------|--------------------------------|---------------------------------|
| <i>RemoteModel</i>      | −0.014<br>(0.043)                | 0.021<br>(0.055)               | 0.061**<br>(0.025)              |
| Controls                | Yes                              | Yes                            | Yes                             |
| CHC F.E.s               | Yes                              | Yes                            | Yes                             |
| Year F.E.s              | Yes                              | Yes                            | Yes                             |
| Observations            | 1483                             | 1483                           | 1292                            |
| Adjusted R <sup>2</sup> | 0.897                            | 0.874                          | 0.644                           |

*Notes:* Time-varying controls include the local mental health specialist-to-population ratio, a prior-year top-quality indicator, the percentage of Medicaid patients, the percentage of homeless patients, indicators for provision of pediatric and dental services, electronic health record adoption, telehealth use in other service lines, rural location, number of service sites, receipt of behavioral health grants, and the local percentage of households with computing device access. Standard errors are clustered at the CHC level. F.E.s = fixed effects.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 7: Specialist-Facilitated Telehealth and Search Frictions

|                                                               | (1)<br><i>Ln(MentalPatients)</i> | (2)<br><i>Ln(MentalVisits)</i> |
|---------------------------------------------------------------|----------------------------------|--------------------------------|
| <i>SpecialistFacilitatedModel</i>                             | −0.002<br>(0.075)                | 0.028<br>(0.075)               |
| <i>SpecialistFacilitatedModel</i><br>× <i>HighSearchCosts</i> | 0.175**<br>(0.081)               | 0.229**<br>(0.104)             |
| Controls                                                      | Yes                              | Yes                            |
| CHC F.E.s                                                     | Yes                              | Yes                            |
| Year F.E.s                                                    | Yes                              | Yes                            |
| Observations                                                  | 1167                             | 1167                           |
| Adjusted R <sup>2</sup>                                       | 0.879                            | 0.853                          |

*Notes:* *HighSearchCosts* is a binary indicator equal to one if the relevant local specialist-to-population ratio is below the sample median, and zero otherwise. Time-varying controls include a prior-year top-quality indicator, the percentage of Medicaid patients, the percentage of homeless patients, indicators for provision of pediatric and dental services, electronic health record adoption, telehealth use in other service lines, rural location, number of service sites, receipt of behavioral health grants, and the local percentage of households with computing device access. Standard errors are clustered at the CHC level and are reported in parentheses. F.E.s = fixed effects.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 8: Remote Telehealth and Setup Frictions

|                                               | (1)<br><i>Ln(MentalPatients)</i> | (2)<br><i>Ln(MentalVisits)</i> |
|-----------------------------------------------|----------------------------------|--------------------------------|
| <i>RemoteModel</i>                            | 0.094*<br>(0.052)                | 0.131**<br>(0.065)             |
| <i>RemoteModel</i> ×<br><i>HighSetupCosts</i> | −0.204**<br>(0.081)              | −0.206**<br>(0.102)            |
| Controls                                      | Yes                              | Yes                            |
| CHC F.E.s                                     | Yes                              | Yes                            |
| Year F.E.s                                    | Yes                              | Yes                            |
| Observations                                  | 1483                             | 1483                           |
| Adjusted R <sup>2</sup>                       | 0.898                            | 0.875                          |

*Notes:* *HighSetupCosts* is a binary indicator equal to one if the share of households with a computing device (e.g., desktop, laptop, smartphone, or tablet) in the CHC service area is below the sample median, and zero otherwise. Time-varying controls include the local mental health specialist-to-population ratio, a prior-year top-quality indicator, the percentage of Medicaid patients, the percentage of homeless patients, indicators for provision of pediatric and dental services, electronic health record adoption, telehealth use in other service lines, rural location, number of service sites, and the receipt of behavioral health grants. Standard errors are clustered at the CHC level and are reported in parentheses. F.E.s = fixed effects.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 9: Remote Telehealth and Transportation Frictions

|                                                        | (1)<br><i>Ln(MentalPatients)</i> | (2)<br><i>Ln(MentalVisits)</i> |
|--------------------------------------------------------|----------------------------------|--------------------------------|
| <i>RemoteModel</i>                                     | −0.031<br>(0.053)                | −0.039<br>(0.060)              |
| <i>RemoteModel</i> ×<br><i>HighTransportationCosts</i> | 0.075<br>(0.077)                 | 0.186*<br>(0.099)              |
| Controls                                               | Yes                              | Yes                            |
| CHC F.E.s                                              | Yes                              | Yes                            |
| Year F.E.s                                             | Yes                              | Yes                            |
| Observations                                           | 1408                             | 1408                           |
| Adjusted R <sup>2</sup>                                | 0.897                            | 0.871                          |

*Notes:* *HighTransportationCosts* is a binary indicator equal to one if the share of households without a vehicle in the CHC service area is above the median, and zero otherwise. Time-varying controls include the local mental health specialist-to-population ratio, a prior-year top-quality indicator, the percentage of Medicaid patients, the percentage of homeless patients, indicators for provision of pediatric and dental services, electronic health record adoption, telehealth use in other service lines, rural location, number of service sites, the receipt of behavioral health grants, and the local percentage of households with computing device access. Standard errors are clustered at the CHC level and are reported in parentheses. F.E.s = fixed effects.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 10: Remote Telehealth and Tracking Frictions

|                                                  | (1)<br><i>ProtocolAdherence</i> |
|--------------------------------------------------|---------------------------------|
| <i>RemoteModel</i>                               | 0.010<br>(0.037)                |
| <i>RemoteModel</i> ×<br><i>HighTrackingCosts</i> | 0.085*<br>(0.043)               |
| Controls                                         | Yes                             |
| CHC F.E.s                                        | Yes                             |
| Year F.E.s                                       | Yes                             |
| Observations                                     | 1292                            |
| Adjusted R <sup>2</sup>                          | 0.644                           |

*Notes:* *HighTrackingCosts* is a binary indicator equal to one if the share of missing income data in the CHC is above the median, and zero otherwise. Time-varying controls include the local mental health specialist-to-population ratio, a prior-year top-quality indicator, the percentage of Medicaid patients, the percentage of homeless patients, indicators for provision of pediatric and dental services, electronic health record adoption, telehealth use in other service lines, rural location, number of service sites, receipt of behavioral health grants, and the local percentage of households with computing device access. Standard errors are clustered at the CHC level and are reported in parentheses. F.E.s = fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 11: Summary of Robustness Checks

| Analysis                                                    | Description                                                                                                                                     | Key Insights                                                                                                                                                                              |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event study analysis of pre-trends                          | We estimate relative time specifications with leads and lags of adoption.                                                                       | Pre-treatment lead coefficients are jointly insignificant, supporting the parallel trends assumption.                                                                                     |
| Stacked difference-in-differences                           | We implement a stacked DiD design that compares each treated cohort to never-treated controls, avoiding comparisons with already-treated units. | Results are consistent with the baseline specification, mitigating concerns about bias from staggered adoption timing.                                                                    |
| Instrumental variable and control function specification    | We estimate a selection-corrected model (Heckman style) using external instruments for telehealth adoption.                                     | Estimated treatment effects remain similar in magnitude and significance, indicating robustness to potential self-selection into adoption.                                                |
| Placebo tests with randomized adoption timing               | We randomly reassign adoption years across CHCs while preserving the annual number of adopters and re-estimate the main model.                  | The observed treatment effects lie in the extreme upper tail of the placebo distribution, suggesting that results are unlikely to be driven by endogenous timing.                         |
| Decomposition of treatment effect variation (Goodman-Bacon) | We decompose the two-way fixed effects DiD estimator into pairwise comparisons following Goodman-Bacon (2021).                                  | The majority of identifying weight comes from treated vs. never-treated comparisons; negative-weight components are limited and do not drive the results.                                 |
| Alternative outcome distributions (count data models)       | We re-estimate access outcomes using PPML and Negative Binomial models with fixed effects.                                                      | Results are consistent with the log-linear specifications, indicating robustness to functional-form assumptions.                                                                          |
| Organizational scale adjustment                             | We control for log(total patients) or log(total visits) to flexibly account for CHC size.                                                       | The specialist-facilitated effect remains positive and significant after accounting for organizational scale, indicating that the findings are not mechanically driven by overall growth. |
| Robustness to outliers (winsorized outcomes)                | We winsorize dependent variables at conventional thresholds to limit the influence of extreme observations.                                     | Main coefficients remain similar, suggesting that the findings are not driven by outliers.                                                                                                |
| Alternative treatment coding assumptions                    | We test sensitivity to alternative coding of partially observed 2017 treatment indicators, using lower- and upper-bound imputations.            | Coefficients are virtually unchanged, suggesting that identification does not hinge on first-year treatment classification.                                                               |
| Placebo outcome tests (domain specificity)                  | We re-estimate the baseline DiD using outcomes that should not be affected by telehealth adoption.                                              | Telehealth adoption is not significantly associated with placebo outcomes, reinforcing that the main effects are domain-specific.                                                         |